

Francesco Cavina

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Data and AI engineer with a BSc in Econometrics and Data Science from VU Amsterdam and incoming MSc studies in Econometrics and Operations Research and Business Analytics. Production experience building data pipelines, local LLM infrastructure, AI ready data interfaces and full stack prototypes. Strong fit for roles that turn analytical models into practical software for operational decision making.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL
- **AI and Data:** LLMs, RAG, ChromaDB, OpenAI API, Hugging Face, scikit learn, PySpark
- **Web and Product:** FastAPI, Next.js, React, Tailwind CSS, REST APIs, UI components
- **Infrastructure:** Git, Docker, Kubernetes, Terraform, Azure, Databricks, Airflow
- **Relevant Interests:** data driven products, dashboarding, UX/UI, Gen AI tools, operational modelling

EXPERIENCE

Data & AI Platform Engineer

Robodata
Netherlands

September 2025 - June 2026, Amsterdam,

- Built privacy aware AI and data platform components for enterprise clients, including metadata driven ingestion, schema inference and PII classification workflows.
- Developed automated data pipelines in Python, PySpark and SQL using Bronze, Silver and Gold architecture.
- Built local LLM workflows for data discovery, field classification and metadata generation.
- Created AI ready data interfaces that allow governed datasets to be queried safely by LLM agents.
- Worked on self service data tooling, making technical data infrastructure easier for users to understand and operate.
- Used Docker, Kubernetes, Terraform, Azure, Airflow and SQLAlchemy across production platform work.

Operational Trader

QuantFi
Netherlands

October 2022 - May 2023, Amsterdam,

- Developed and tested ML based trading strategies in Python using Random Forest, XGBoost, neural networks and walk forward validation.
- Built data ingestion, retraining and daily forecast workflows using Python and SQL.
- Created performance dashboards and visualisations to evaluate model output and support daily decisions.

Teaching Assistant

Vrije Universiteit Amsterdam
Netherlands

June 2022 - November 2024, Amsterdam,

- Taught weekly Probability Theory tutorials to first year Econometrics students.
- Explained abstract mathematical concepts in a clear and practical way.
- Built a student portfolio system for first year students and represented the programme during Open Days.

EDUCATION

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 •
Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds
• Leeds, England • 2024 • 8.5/10

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making • Vrije Universiteit Amsterdam
• Amsterdam, Netherlands • 2026-2029

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam
• Amsterdam, Netherlands • 2026-2029

PROJECTS

- 1) [AI Powered Personal Website](https://github.com/FrancescoCavina02/My-Website) • <https://github.com/FrancescoCavina02/My-Website> • November 2025 - present
Built a full stack portfolio with a FastAPI backend that parses an Obsidian vault into a REST API for notes, quotes and chat functionality. Used Next.js, TypeScript, Python and Tailwind CSS.
- 2) [RAG Knowledge Chatbot](https://github.com/FrancescoCavina02/Spiritual-chatbot) • <https://github.com/FrancescoCavina02/Spiritual-chatbot> • November 2025 - present
Built a retrieval augmented chatbot using semantic chunking, vector search, structured prompts and citation based responses. Used Python, FastAPI, ChromaDB, sentence transformers and LLM APIs.
- 3) [Dodgeball Club Amsterdam](https://github.com/FrancescoCavina02/DCA-website) • <https://github.com/FrancescoCavina02/DCA-website> • June 2025 - present
Built and maintain a public website using HTML, CSS and JavaScript. Focused on clear navigation, simple content management and practical usability for club members.

Podcasts

[Podcast Link](#) - Created two independent interview podcasts with researchers in physics, cosmology, mathematics and philosophy.